Friday Flex — Chapter 5: Collisions Transfer Energy

> Friday = FLEX only. No essential new content. Nothing here is required for Monday.

RECOVER (finish + catch up)

- **Finish Crash Lab data** — complete Part A / Part B displacement tables and the compare lines.
- **Notebook catch-up** — Insert 1 (collision vocab) · Insert 2 (collision model + fair-test rule) · Insert 3 (misconception flap).
- **Reader recovery** — reread Ch.5 pp. 25–30 (motion / shape / sound / speed / weight).

PRACTICE — Collision Predictor (CUSTOM)

- **Time:** 10–15 min · **Materials:** printed cards / worksheet · **Independence:** guided
- Given a mini-scenario (same car, higher ramp; same ramp, heavier car; faster car vs slower car), students predict which collision causes a greater change and circle the ONE variable that changed.
- **Concepts reinforced:** variable isolation, prediction from prior observation.

PRACTICE — Fair-Test Detective (CUSTOM)

- **Time:** 10 min · **Materials:** worksheet · **Independence:** independent
- Given 5 flawed mini-plans (e.g., "change the car AND the ramp at once"), students circle what was changed and fix the plan to change ONE thing.
- **Concepts reinforced:** one-variable-at-a-time.

PRACTICE — Cause → Effect Matching (CUSTOM)

- **Time:** 10 min · **Materials:** cards · **Independence:** independent
- Match CAUSE (faster car · heavier car · slower car · lighter car) to EFFECT (target moves farthest · next · less · least).
- **Concepts reinforced:** speed/weight → change relationship.

EXTEND — Newton's Cradle Discussion (EXTENSION)

- **Time:** 10 min · **Materials:** Newton's cradle (teacher demo) · **Independence:** guided
- Watch the cradle; discuss: "Where did the motion energy go?" (through the balls and

back). Reinforce: energy transfers, never disappears.

- **Concepts reinforced:** energy transfer, energy does not disappear.

EXTEND — Everyday Collision Hunt (EXTENSION)

- **Time:** 15 min · **Materials:** printed list · **Independence:** independent
- Find collisions at home/school (ball → bat, hammer → nail, foot → soccer ball). For each: what changed? where did the energy go?
- **Concepts reinforced:** collision changes (motion/shape/sound), energy transfer.

Rule check

- [x] Not required for Monday's new core instruction
- [x] Low-prep, quick reset (end-of-day prep)
- [x] No heat measurement · no force/momentum vocabulary · Newton's cradle = demo only